

21. (a) Berkeley and Hartley's method

22. (a) Isotonic

23. (d) $P/C = RT$

Explanation:

$$\pi = CRT$$

So,

$$\frac{\pi}{C} = RT$$

24. (d) All of the above

25. (a) Sugar beet will lose water from its cells

Explanation:

Concentrated NaCl solution is hypertonic.

26. (b) $\pi V = nRT$

$$\pi V = nRT$$

27. (c) $P \propto V$ if T constant

28. (b) Equal osmotic pressure

29. : (b) Same weight concentration

30. (d) None of these

31. Explanation:

0.6% urea:

$$\frac{0.6}{60} = 0.01 \text{ mol}$$

in 100 mL $\rightarrow 0.1M$

Glucose is non-electrolyte, so same molarity gives same osmotic pressure.



(a) 0.1 M glucose

32. Using:

$$\pi = CRT$$

$$\pi = 0.2 \times 0.082 \times 293$$

$$\pi \approx 4.8 \text{ atm}$$

(c) 4.8 atm

33. (b) Osmosis

34. (d) Isotonic

35. (b) Solutions of non-electrolytes only

36. (c) Saturated solution of CO_2
Explanation: CO_2 escapes from solution.

37. (c) Lower than theoretical
Explanation:
NaCl dissociates, increasing particle number.

38. (c) Number of solute molecules increases

39. (d)

Explanation:

For sucrose:

$$\frac{3.42}{342} = 0.01M$$

For NaCl:

$$\frac{1.17}{58.5} = 0.02M$$

Since NaCl dissociates into 2 ions:

$$iM = 2 \times 0.01 = 0.02$$





Both have same effective concentration.

40. (d) Directly proportional to absolute temperature

Explanation:

$$\pi = CRT$$

Hence osmotic pressure is directly proportional to absolute temperature.

